

23 The Evolution of Populations

KEY CONCEPTS

- 23.1** Genetic variation makes evolution possible *p. 487*
- 23.2** The Hardy-Weinberg equation can be used to test whether a population is evolving *p. 489*
- 23.3** Natural selection, genetic drift, and gene flow can alter allele frequencies in a population *p. 493*
- 23.4** Natural selection is the only mechanism that consistently causes adaptive evolution *p. 497*

Study Tip

Make a table: This chapter discusses different types of selection—disruptive, stabilizing, sexual, balancing, frequency-dependent, and heterozygote advantage. These types of selection are not all mutually exclusive, and some evolutionary changes are examples of more than one. To help you study, make a table like the one shown listing each type of selection, its distinguishing features, and example(s) from the chapter.

Type of selection	Distinguishing features	Example
Balancing selection	Maintains two or more phenotypes in a population; includes frequency-dependent selection and heterozygote advantage	Figures 23.17 and 23.18
Directional selection		

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For Students (in eText and Study Area)

- Get Ready for Chapter 23
- Figure 23.8 Walkthrough: Hardy-Weinberg Equilibrium
- BioFlix® Animation: Natural Selection, Genetic Drift, and Gene Flow

For Instructors to Assign (in Item Library)

- Tutorial: The Hardy-Weinberg Principle

Ready-to-Go Teaching Module

- (in Instructor Resources)
- Mechanisms of Evolution



Figure 23.1 The medium ground finch (*Geospiza fortis*) is a seed-eating bird that inhabits the Galápagos Islands. In 1977, the *G. fortis* population on the island of Daphne Major was decimated by a long period of drought: Of some 1,200 birds, only 180 survived. The surviving birds had larger, deeper beaks, indicating that this population of finches had evolved.

What mechanisms can cause the evolution of populations?

Evolution occurs when natural selection, genetic drift, or gene flow alter the allele frequencies of a population over time.

